# Centering the error makes the posterior mean optimal

**Average the centered expansion** Previous slide: pointwise squared-error expansion

$$\|z-a\|^2 = \|r\|^2 + 2r^T(m-a) + \|m-a\|^2$$

**Take $E[\cdot\,|y]$; $m-a$ is fixed**

$$E[r|y] = 0$$

$$E\left[r^T(m-a)|y\right] = E[r|y]^T(m-a) = 0$$

**With $V = \mathrm{Cov}[z|y] = E[rr^T|y]$**

$$E\left[\|r\|^2|y\right] = \sum_{j=1}^{d} E\left[r_j^2|y\right] = \operatorname{tr} V$$

**Conditional squared-error risk**

$$R(a|y) = \operatorname{tr} V + \|m-a\|^2$$

**Optimal reconstruction**

$$\|m-a\|^2 \geq 0 \quad \text{with equality iff} \quad a = m$$

$$a^*(y) = m = E[z|y]$$

$$\min_a R(a|y) = \operatorname{tr} V$$

Gaussian: $a^*(y) = \mu_{z|y}, \quad \min_a R(a|y) = \operatorname{tr} \Sigma_{z|y}$

- Risk is residual uncertainty plus distance from the mean.
- The conditional mean minimizes model expected squared error.